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Step Doubling for Pipeline Flow

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ABSTRACT

This paper defines and studies a simple, efficient method for discretizing pipeline equations in time.

In many ways the solutions of pipe flow equations look like the solutions of nonlinear diffusion equations, so it is natural that schemes that work well for diffusion problems might have analogs for pipe flow problems.

The method presented here is based on backward Euler (BE). To advance the solution from time t to time $t + dt$ it uses a single step of BE with step size dt and two steps of BE with step size $dt/2$. The results are then combined to give a second-order correct scheme. This process is called step doubling with local extrapolation (SDoLE) and has been rigorously analyzed for nonlinear diffusion problems in the context of Galerkin spatial discretizations. [1] Here a collocation scheme is the underlying discretization in space and time.

The backward Euler that forms the basis of this method is done using a linearization technique instead of solving a nonlinear system. This is extended to minimize the computational costs associated with evaluation of the nonlinearities in the equations. The resulting discretization technique is called affine step doubling with local extrapolation (ASDoLE).

The properties of the method are explained by looking first at a scalar ODE and then by presenting examples based on

pipeline operation. A single pipe case is given followed by a simple gun barrel example.

INTRODUCTION AND BACKGROUND

Simulating the behavior of pipeline networks has challenges that are associated with the fact that control changes are often made on the time scale of a few minutes while the results needed cover hours or days. Here we adapt a technique that has been found useful in the context of diffusion problems to studying flow in pipeline networks. Selecting a time integration method involves a balance between accuracy, efficiency, and simplicity; those considerations are involved in the examination of the method studied here.

STEP DOUBLING FOR ORDINARY DIFFERENTIAL EQUATIONS

In this section we consider some methods for the approximate solution of the initial value problem

$$\begin{aligned} y'(t) &= F(y) \text{ for } t > 0, \\ y(0) &= y_0. \end{aligned}$$

Here y is an n -vector and F is a well-behaved function.

Euler method

Let Y denote an approximation to y at time t . To advance time by an amount $dt > 0$ using the Euler method we can compute dY , the change in Y over the step, as follows:

$$dY = dt F(Y).$$

This is perhaps the simplest method.

θ -method

For a given parameter θ we can define a θ -method by choosing dY to satisfy

$$dY = dt F(Y + \theta dY).$$

Under reasonable assumptions on F and dt this system will have a unique solution. Note that for $\theta = 0$ this is just the Euler method and for $\theta = 1$ it is the backward difference (or backward Euler) method. For $\theta = 1/2$ this method is second-

order correct and is frequently called the trapezoidal method. When using a θ -method, it is often useful to take θ to be slightly larger than $1/2$, say 0.6 or 0.7 , for improved stability at some loss in accuracy.

If F is nonlinear and θ is nonzero the above θ -method requires that we solve a nonlinear system at each time step. In these situations, it may be useful to use an affine θ -method:

$$dY = dt(F(Y) + \theta DF(Y)dY),$$

where $DF(Y)$ is the derivative of F evaluated at Y . In this case we need only solve a linear system for dY .

SDoLE & ASDoLE methods

To have a second-order correct method that has good stability properties we can build a step doubling approximation using three different backward difference steps. Let

$$\begin{aligned} dS &= dt F(Y + dS), \\ S &= Y + dS, \\ dD_{1/2} &= (dt/2) F(Y + dD_{1/2}), \\ D_{1/2} &= Y + dD_{1/2}, \\ dD_1 &= (dt/2) F(D_{1/2} + dD_1), \\ D &= D_{1/2} + dD_1, \\ Y &= 2D - S. \end{aligned}$$

Note that S and D are both approximations to Y at $t + dt$; S comes from a single step of backward difference and D comes from two half steps of backward difference. The error over this step in D should be about half of the error in S , so the formula for updating Y is second-order correct. Since this method involves both step doubling and local extrapolation, a reasonable acronym for it is SDoLE, Step Doubling with Local Extrapolation.

The methods described here have “step doubling” in their names. This goes back at least to C.W. Gear’s 1971 book on stiff differential problems [2]. There it was recommended for error estimation and step size control. The use of local extrapolation to improve accuracy is frequently recommended in the literature for Runge Kutta methods.

To avoid the solution of nonlinear equations, F above can be replaced by an affine approximation on the time interval $[t, t + dt]$:

$$F(Y + z) \approx F(Y) + DF(Y)z.$$

This then gives

$$\begin{aligned} dS &= dt (F + DF dS), \\ S &= Y + dS, \\ dD_{1/2} &= (dt/2)(F + DF dD_{1/2}), \\ D_{1/2} &= Y + dD_{1/2}, \\ dD_1 &= (dt/2)(F + DF(D_{1/2} + dD_1)), \\ D &= D_{1/2} + dD_1, \\ Y &= 2D - S, \end{aligned}$$

where we used F and DF for $F(Y)$ and $DF(Y)$, respectively. This allows us to update Y by only solving linear systems.

This method will be referred to ASDoLE, an acronym for Affine Step Doubling with Local Extrapolation.

It is of interest that in advancing time by dt we only evaluated F and DF once.

Scalar case

To build our intuition about these methods it is helpful to consider the scalar case where F is linear. Consider

$$y'(t) = -ay(t).$$

Think of a as being positive. Then the solution decays as t increases. The solution has the property

$$y(t + dt) = \exp(-a dt)y(t).$$

If Y is the solution of the θ -method, then

$$Y(t + dt) = \frac{1 - (1 - \theta)a dt}{1 + \theta a dt} Y(t).$$

For Y given by SDoLE or ASDoLE we get

$$Y(t + dt) = \left[2 \left(\frac{1}{1 + a dt/2} \right)^2 - \frac{1}{1 + a dt} \right] Y(t)$$

or

$$Y(t + dt) = A(a dt)Y(t)$$

where

$$A(a dt) = \left[2 \left(\frac{1}{1 + a dt/2} \right)^2 - \frac{1}{1 + a dt} \right].$$

In each step, going from t to $t + dt$ is just multiplication by an “amplification factor”, $A(a dt)$. Below in figure 1 are the “amplifications factors” for the exact solution, step doubling (SD), and the θ -method for $\theta = 0.6$ which is the default in SPS.

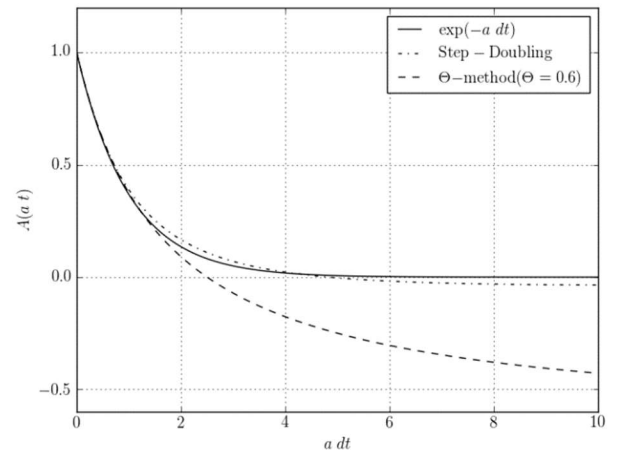


Fig. 1 The amplification factors for the exact solution, SD, and the θ -method. Note that SD behaves as a rational approximation to the exponential function for a linear equation which is the exact solution in this case.

The properties of the amplification factors tell us something about how the time-stepping behaves when there’s a perturbation in the initial conditions. To see this behavior for

this scalar linear problem, consider two different steps, which are similar except for a small difference in the prior step, ϵ :

$$\begin{aligned} y_1 &= A(a \, dt)y_0 \\ z_1 &= A(a \, dt)z_0 = A(a \, dt)(y_0 + \epsilon) \end{aligned}$$

The difference between the steps is

$$d_1 = z_1 - y_1 = A(a \, dt)(z_0 - y_0) = A(a \, dt)\epsilon.$$

This can be generalized, allowing us to see how the difference in an initial step can influence future steps,

$$d_i = A^i(a \, dt)\epsilon, \quad i = 1, 2, 3 \dots$$

The following are some observations regarding how this difference in a prior step can perpetuate to subsequent steps: If,

- $1 < A(a \, dt)$, the difference will grow;
- $A(a \, dt) = 1$, the difference will be constant;
- $0 < A(a \, dt) < 1$, the difference will shrink monotonically;
- $-1 < A(a \, dt) < 0$, the difference will shrink but oscillate;
- $A(a \, dt) = -1$, the difference will oscillate;
- $A(a \, dt) < -1$, the difference will oscillate and grow.

STEP DOUBLING FOR A SINGLE PIPE

In this section the use of step doubling is extended to the case of a single pipe. Attention will be restricted to isothermal gas pipeline flow with pressures given at the pipe ends.

Equations for isothermal gas pipeline flow

For $0 < x < L$ and $t > 0$

$$\begin{aligned} \partial_t p + v \partial_x p + \frac{\rho}{\rho'} \partial_x v &= 0, \\ \partial_t v + \frac{C_0}{\rho} \partial_x p + v \partial_x v + \frac{f}{2d} |v|v &= 0. \end{aligned}$$

Here $p(x, t)$ is pressure, $v(x, t)$ is velocity, C_0 is a units constant, $f(\text{Re})$ is the friction factor, and $\rho(p)$ is the density.

The boundary conditions are taken to be

$$p(0, t) = p_0(t), \quad p(L, t) = p_L(t),$$

where p_0 and p_L are given. These equations can be derived from conservation of mass and momentum, assuming plug-like flow.

Box scheme

The “box scheme” for this system uses a tensor product of continuous piecewise linear functions on $[0, L]$ and $[0, T]$. If the spatial mesh is $\{x_i\}$ and the temporal mesh is $\{t_k\}$ then at the center of each box, $[x_{i-1}, x_i] \times [t_{k-1}, t_k]$, we solve the PDE. The boundary conditions are imposed at the ends of the spatial mesh, at the time levels, t_k .

Luskin [3] analyzed a linearized version of a close variant of this system for networks of pipes with rather general boundary conditions.

For $1/2 \leq \theta \leq 1$, a θ -weighted box scheme uses the mid-

point in x , and $t_\theta = (1 - \theta)t_{k-1} + \theta t_k$ for the collocation point. The scheme is more robust with $\theta > 1/2$, but it is only first order correct. For $\theta = 1$ this is backward difference.

The box scheme, as described here, does not perform well when the CFL number is substantially less than one; the CFL number is $c \, dt/dx$, where c is the speed of sound, dt is the time step, and dx is the space interval length. This behavior is unusual for numerical methods for systems of equations like the ones we consider. For time steps that are so small that a disturbance can only propagate (physically) less than dx in a single step, the scheme needs to be modified, perhaps by decreasing dx . This type of situation comes up when dx is poorly chosen or some part of the pipeline network requires a very small time step. In the examples included in this paper the CFL number is always greater than 1.

ASDole for a pipe

Note that the box scheme discretizes both space and time and it is not suitable for a “method of lines” approach unless it is modified. However, for given θ , dt and x -mesh, it defines the changes in the approximations of p and v just as the θ -method did for the ordinary differential equations. The nonlinear terms can be expressed in terms of their values at the start of a step plus terms involving the derivatives. Since the equations themselves involve a derivative of p , the linearizations use the second derivative of p and the first derivative of f .

Example

To demonstrate the difference in stability between step doubling and the θ -method we use a simple one pipe model. For this case, the pipe is 50 miles (80.5 km) long with an ID of 32.0 inches (81.3 cm). For friction calculations, the Swamee-Jain approximation to the Colebrook equation is used. The EOS is CNGA with a specific gravity of 0.6.

The pipe is initially in steady state with an upstream pressure of 700 psig (4826.3 kPa gauge) and downstream flow of 461.2 mmscf/d (541,950.0 scm/hr). At time 60 minutes, the pressure on the upstream side changes to 900 psig (6205.3 kPa gauge) over a 2 minute time interval. Shortly afterwards at $t=66.5$ min the downstream flow rate is ramped upwards to 700 mmscf/d (284,020.0 scm/hr) by time $t=75.5$ min.

For this numerical demonstration, $dx = 1$ mile (1.6 km) for both the θ -method and SD method. In the time interval $[0, 60.5)$ minutes the time step is 30 seconds to let the method capture the pressure rise. For the remainder of the simulation, $t \geq 60.5$ minutes, the time step is increased to 90 seconds. To compare the accuracy and stability of our trial methods, a finely resolved case was run with a spatial step of 0.05mi (0.08 km) and a fixed time step of 4.8sec. The resolved case is close enough to the true solution for us to quickly see some benefits to using the SD method. The resolved solution could be done with either method as both will converge sufficiently.

Below in figure 2 we can see that the simulated flow into the pipe for the θ -method ($\theta = 0.6$) begins to oscillate upon missing the flow peak of the resolved solution. Step doubling captures this peak more closely in value then settles in closely to the resolved solution without any bounce.

In figure 3 we can see that the θ -method over-estimates the pressure swell on the discharge end of the pipe where step doubling hugs the resolved solution a little more tightly.

A Δt of 90 seconds may be a bit large, but it demonstrates the benefits clearly.

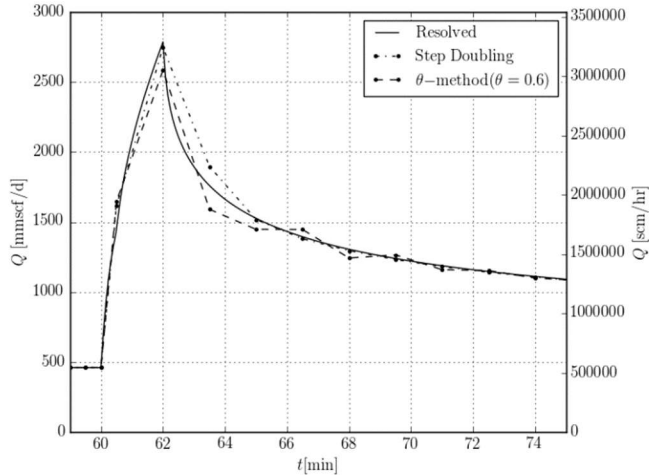


Fig. 2 Input flow for the one pipe example where $\Delta t=90\text{sec}$ after $t=60.5\text{min}$. Oscillation is present in the θ -method solution.

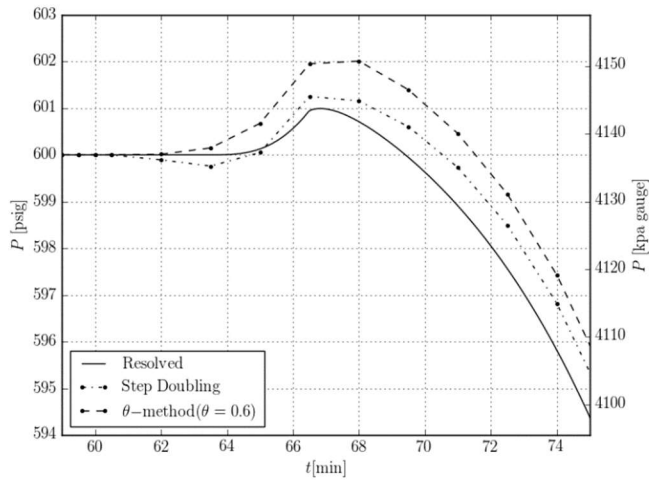


Fig. 3 Output pressure for the one pipe example where $\Delta t=90\text{sec}$ after $t=60.5\text{min}$. The θ -method overshoots the pressure swell.

To get a feel for how the methods perform with a more reasonable time step, we duplicate the experiment above but with a constant time step of 30 seconds. The oscillation in the θ -method is still present but very slight as we see in the plot of the input flow (figure 4). For the input pressure we see that SD converged much closer to the resolved solution than the θ -method (figure 5) for this decrease in time step, but the θ -method still gives a good solution. We can draw the

conclusion from this pair of tests that SD does a better job handling under-resolved time meshes but for an appropriate time step the θ -method can still perform well.

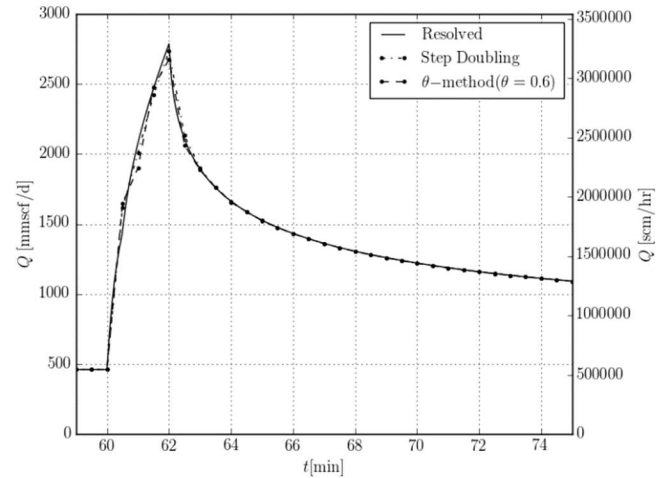


Fig. 4 Input flow for the one pipe example where $\Delta t=30\text{sec}$ for the entire simulation. Oscillation is very slight in the θ -method solution.

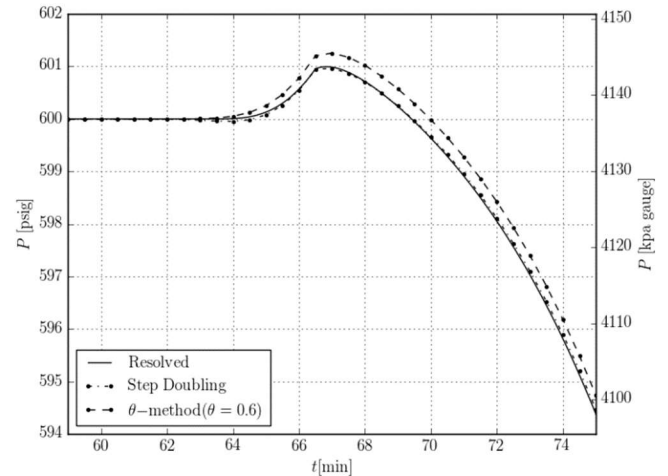


Fig. 5 Output pressure for the one pipe example where $\Delta t=30\text{sec}$ for the entire simulation. Step doubling converges more rapidly as the time step shrinks than the θ -method.

STEP DOUBLING FOR A PIPELINE NETWORK

The ability to simulate the behavior of a single pipe is not usually all that is needed to address the engineering questions that arise from pipeline system design and operation. A fully functional pipeline simulator will in general need to simulate various types of compressors, pressure regulators, valves, headers, PID controllers, scheduled supplies and deliveries, etc. The techniques that are described here for pipes can be incorporated into such a network simulator by only changing the behavior of the pipes.

In the case of a single pipe with pipe-end pressures given, we

can use the boundary data in computing the $D_{1/2}$ state of the pressure and flow in the pipe. In the case of a network simulator that computes the state of the whole network by repeatedly advancing time from t to $t + dt$, the pipe end pressures are usually computed at $t + dt$ by solving a large, sparse set of equations. The pipe-end values for each pipe of the pipe-states denoted by S and D will be required to match these computed pipe-end pressures at $t + dt$. For each pipe-end, the pressures for $D_{1/2}$ will be the average of the pipe-end pressures at t and $t + dt$. In our implementation of this method, we treat this efficiently using a sparse matrix technique known as “tearing”.

Example

To have a more realistic example, we use a three pipe, three compressor gun-barrel model.

All the pipes are 50 miles (80.5 km) long with an ID of 32.0 inches (81.3 cm) as in the previous single pipe example. Between each adjoining pair of pipes is an idealized compressor. The behavior of an idealized compressors is defined by a given time schedule of its discharge pressure. The gas used in this example is close to that in the single pipe example.

For this case, the system is initially at steady state with each pipe having a pressure of 700 psia (4826.3 kPa absolute) at its inlet and 600 psia (4136.9 kPa absolute) at its outlet. Starting at time = 0 minutes the inlet pressure to the first pipe is increased to 900 psia (6205.3 kPa absolute) over a 2 minute period, 20% in the first minute and 80% in the second minute. At time = 22 minutes the compressor between pipes 1 and 2 has its discharge pressure raised on the same schedule. Next, at time = 44 minutes the compressor between pipes 2 and 3 has its discharge pressure raised on the same schedule. The outlet pressure of pipe 3 remains at 600 psia for the entire run. At time = 90 minutes the system is still not in steady state; the inlet flow is still about 10% more than the outlet flow.

The dx in this system is 1 mile (1.6 km) and the dt is 1 minute during the 2 minute periods in which the pipe inlet pressures are changing, otherwise dt is 2 minutes. This rather coarse time stepping allows us to see the effect of the choice of method. As before, a finely resolved case was run for comparison.

Some results are presented in the following figures. We can see that when the compressors “kick on” the step doubling’s solution is closer to the true solution than the θ -method’s. Here $\theta = 0.6$. The θ -method solution even shows some oscillation in the flows as was predicted by the scalar stability analysis.

Note that the pressure difference between the θ - and step doubling-methods in the outlet pressure of pipe 1 in figure 6 is about 15 psi (103.4 kPa) and this persists for quite a while.

This says that the pressure drop is about 5% different, and this will be associated with about a 2.5% difference in flow rate.

While our purpose in this paper is to show some advantage for the step doubling scheme, we note that if we decrease the time steps in this example by a factor of three then the step doubling and θ -method results are very close to those of the finely resolved case. That is to say, the θ -method is a perfectly reasonable method when used with appropriate time step selection. The same remarks apply to the single pipe example presented earlier.

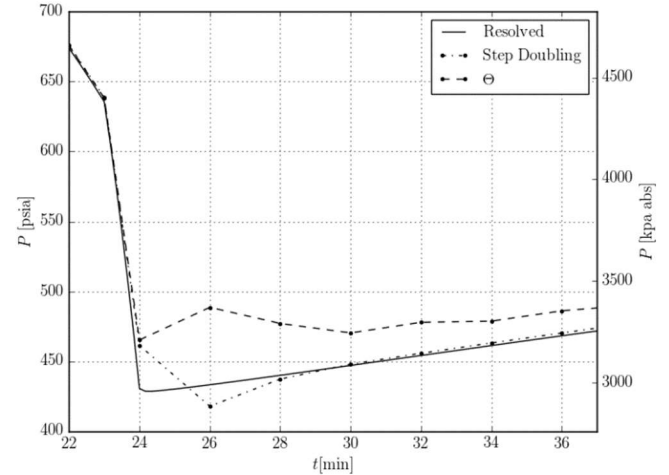


Fig. 6 We can see a large mismatch in pressure solutions at $t=26$ min for the output pressure of pipe 1.

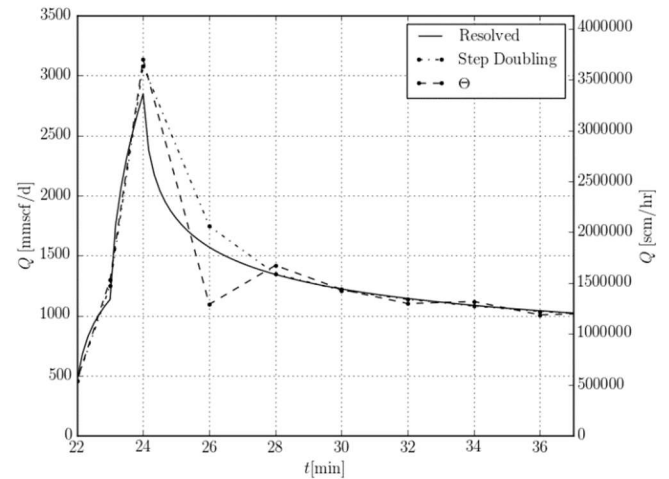


Fig. 7 Both methods over-shoot the peak in flow at $t=24$ min for the output flow of pipe 1. The θ -method bounces around the resolved solution while SD does not.

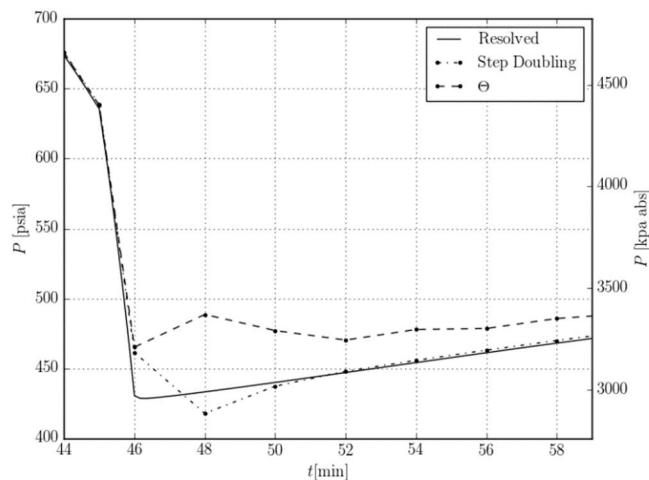


Fig. 8 The output pressure for pipe 2 has similar behavior as the output pressure for pipe 1. The time axis shift is due to the delayed setpoint rise of compressor 1.

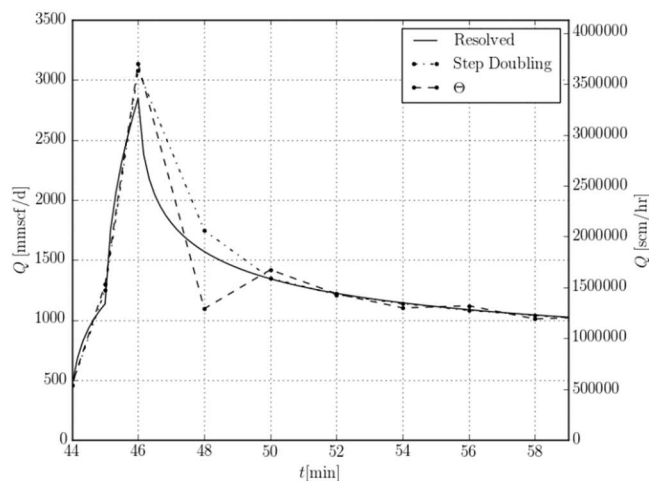


Fig. 9 We get a similarly bouncy solution for pipe 2's output flow.

CONCLUSIONS

Step doubling(SD) handles coarser time meshes better than the θ -method, but θ -method is still very good when the appropriate time mesh is used.

Experiments and analysis show that SD is less oscillatory than the θ -method.

Amplification factors indicate that the behavior of SD is closer than the θ -method to the exact solution and that numerical disturbances accumulate less.

In the case of linear equations, this step doubling method is a rational approximation to the exponential, i.e. the exact solution, given by an exponential, is approximated using the ratio of two polynomials, a quadratic divided by a cubic.

Investigation of various rules for automatic time step selection is desirable, particularly ones that utilize SD's built in error estimation, the difference between S and D. The goal is to maintain accuracy and improve efficiency by placing computation only where it is needed along the time axis.

The cases presented here were only isothermal, of course the integration with a thermal numerical model is necessary.

REFERENCES

1. B. Ayati and T.F. Dupont, Convergence of a step doubling Galerkin method for parabolic problems, *Math. Comp.* 74 (2005) 1053-1065.
2. C.W. Gear, *Numerical Initial Value Problems in Ordinary Differential Equations*, Prentice-Hall, 1971.
3. M. Luskin, An Approximation Procedure for Nonsymmetric, Nonlinear Hyperbolic Systems with Integral Boundary Conditions. *SIAM Journal on Numerical Analysis*, 1979, Vol. 16, No. 1: pp. 145-164.

AUTHOR BIOGRAPHY

Todd Dupont was a pioneer in the simulation of pipeline flow, an area he has worked in for over 45 years. His work has been presented at several PSIG conferences and he received a PSIG Service Award in 2009. He is a professor at the University of Chicago, whose research interests are primarily in numerical analysis and condensed matter physics. Two of his PhD students wrote theses on simulation of pipelines. Professor Dupont's graduate and undergraduate degrees are from Rice University. He has authored or co-authored over 100 publications for which Google Scholar counts over 15,000 citations.

Richard Myers, Ph.D. is a former mathematics professor from the University of St. Thomas - Houston and is currently a senior software development scientist in DNVGL's Digital Solutions Analytics Advancement group. He joined DNVGL back when it was Advantica in 2007 and has worked on many projects over the years. Most of his time was spent working on Attune and GTO. He has also worked on various features in Synergi Gas and SPS.